



JOHNS HOPKINS

WHITING SCHOOL
of ENGINEERING

MECHANICAL ENGINEERING COURSE PLANNING MAPS

Note: This document is intended to serve as a scheduling aid for Mechanical Engineering students. The schedule information provided is based on historical patterns and expected scheduling. Course offerings and modalities are subject to change due to instructor availability, student demand, and program priority. Students should check SIS for official course schedule information and the Johns Hopkins University Academic Catalog for official degree requirements.

FOCUS AREAS

Click to navigate to your focus area.

ADVANCED MANUFACTURING

HYPersonic TECHNOLOGIES

AEROSPACE ENGINEERING

MECHANICS OF MATERIALS AND
STRUCTURES

BIOMECHANICS

OCEAN ENGINEERING

FLUID AND THERMAL MECHANICS

ROBOTICS, DYNAMICS, AND
CONTROLS

ADVANCED MANUFACTURING

10 Courses: 1 core mathematics course + 2 Group 1 courses + 3 courses from Group 1 or 2 + 4 technical electives

Group 1 Courses

535.628
Computer-Integrated Design &
Manufacturing

Every Term

535.642
Control Systems for Mechanical
Engineering Applications

Fall

535.659
Manufacturing System Analysis

Fall

535.660
Precision Mechanical Design

Fall/Spring

535.673
Mechanized Assembly: Hardware
and Algorithms

Summer

535.674
Smart Manufacturing and
Automation

Spring

535.721
Advanced Composite Materials &
Manufacturing Processes

Spring/Summer

Group 2 Courses

535.737
Multiscale Modeling & Simulation
of Mechanical Systems

Summer

535.684
Modern Polymeric Materials

Fall/Spring

535.720
Mechanics of Composite
Materials and Structures

Fall

535.627
Computer-Aided Design

Every Term

535.744
AI for Mechanical Materials
Design

Spring

Computational

Materials

535.618
Fabricatology: Advanced
Materials Processing

Summer/Fall

535.638
Mechanical Packaging for
Electronics Systems

Summer/Fall

535.733
Micromechanics of
Heterogeneous Materials &
Composites

Fall

535.672
Advanced Manufacturing
Systems

Summer

515.658
Design for Additive
Manufacturing

Fall

515.622
Micro and Nano Structured
Materials & Devices

TBD

Applications

AEROSPACE ENGINEERING [AERODYNAMICS FOCUS AREA]

10 Courses: 1 core mathematics course + 2 Group 1 courses + 3 courses from Group 1 or 2 + 4 technical electives

Group 1 Courses

535.608
Hypersonic Technologies and
Systems

Fall

535.620
Fluid Dynamics I

Summer/Fall

535.626
Mechanics of Flight

Fall/Spring

535.670
Advanced Aerodynamics

Fall

Group 2 Courses

535.608
Hypersonic Technologies and
Systems

Fall

535.623
Intermediate Vibrations

Fall/Spring

535.625
Turbulence

Fall

535.626
Mechanics of Flight

Fall/Spring

535.639
Aerospace Materials

Spring

535.652
Thermal Systems Design and
Analysis

Fall

535.670
Advanced Aerodynamics

Fall

535.735
Computational Fluid Dynamics

Fall

535.752
Advanced Flight Dynamics and
Control of Aerospace Vehicles

Fall

535.761
Hypersonic Aerothermodynamics

Spring

535.763
Aerospace Propulsion

Spring

AEROSPACE ENGINEERING [DYNAMICS & CONTROL FOCUS AREA]

10 Courses: 1 core mathematics course + 2 Group 1 courses + 3 courses from Group 1 or 2 + 4 technical electives

Group 1 Courses

535.608
Hypersonic Technologies and
Systems

Fall

535.626
Mechanics of Flight

Fall/Spring

535.752
Advanced Flight Dynamics and
Control of Aerospace Vehicles

Fall

535.762
Guidance, Navigation and
Controls for Hypersonic Vehicles

Fall/Spring

Group 2 Courses

535.608
Hypersonic Technologies and
Systems

Fall

535.612
Intermediate Dynamics

Fall

535.623
Intermediate Vibrations

Fall/Spring

535.626
Mechanics of Flight

Fall/Spring

535.642
Control Systems for Mechanical
Engineering Applications

Fall

535.670
Advanced Aerodynamics

Fall

535.724
Dynamics of Robots and
Spacecraft

Spring

535.741
Optimal Control and
Reinforcement Learning

Spring

535.752
Advanced Flight Dynamics and
Control of Aerospace Vehicles

Fall

535.762
Guidance, Navigation and
Controls for Hypersonic Vehicles

Fall/Spring

535.763
Aerospace Propulsion

Spring

525.609
Continuous Control Systems

Fall/Spring

525.645
Modern Navigation Systems

Fall/Spring

525.661
UAV Systems and Controls

Fall/Spring

525.777
Control System Design Methods

Fall

AEROSPACE ENGINEERING [STRUCTURES & MATERIALS FOCUS AREA]

10 Courses: 1 core mathematics course + 2 Group 1 courses + 3 courses from Group 1 or 2 + 4 technical electives

Group 1 Courses

535.606 Advanced Strength of Materials Spring
535.607 Mechanics of Solids & Structures: Theory and Applications I Fall
535.632 Applied Finite Elements Every Term

Group 2 Courses

535.606 Advanced Strength of Materials Spring	535.607 Mechanics of Solids & Structures: Theory and Applications I Fall	535.608 Hypersonic Technologies and Systems Fall	535.612 Intermediate Dynamics Fall
535.623 Intermediate Vibrations Fall/Spring	535.627 Computer-Aided Design Every Term	535.628 Computer-Integrated Design & Manufacturing Every Term	535.632 Applied Finite Elements Every Term
535.639 Aerospace Materials Spring	535.643 Plasticity Spring	535.652 Thermal Systems Design and Analysis Fall	535.660 Precision Mechanical Design Fall/Spring
535.684 Modern Polymeric Materials Fall/Spring	535.706 Mechanics of Solids & Structures: Theory and Applications II Spring	535.720 Mechanics of Composite Materials and Structures Fall	535.721 Advanced Composite Materials & Manufacturing Processes Spring/Summer
	535.731 Engineering Materials: Properties and Selections Fall/Spring	535.732 Fatigue and Fracture of Materials Spring	
515.658 Design for Additive Manufacturing Fall	515.661 Introduction to Polymer Science Fall	675.702 Spacecraft Materials Every Term	675.722 Space Mechanical Design and Analysis In Development

AEROSPACE ENGINEERING [VEHICLE DESIGN FOCUS AREA]

10 Courses: 1 core mathematics course + 2 Group 1 courses + 3 courses from Group 1 or 2 + 4 technical electives

Group 1 Courses

535.608
Hypersonic Technologies and
Systems
Fall

535.626
Mechanics of Flight
Fall/Spring

535.752
Advanced Flight Dynamics and
Control of Aerospace Vehicles
Fall

Group 2 Courses

535.606
Advanced Strength of Materials
Spring

535.607
Mechanics of Solids & Structures:
Theory and Applications I
Fall

535.608
Hypersonic Technologies and
Systems
Fall

535.626
Mechanics of Flight
Fall/Spring

535.627
Computer-Aided Design
Every Term

535.628
Computer-Integrated Design &
Manufacturing
Every Term

535.632
Applied Finite Elements
Every Term

535.652
Thermal Systems Design and
Analysis
Fall

535.660
Precision Mechanical Design
Fall/Spring

535.670
Advanced Aerodynamics
Fall

535.706
Mechanics of Solids & Structures:
Theory and Applications II
Spring

535.720
Mechanics of Composite
Materials and Structures
Fall

535.721
Advanced Composite Materials &
Manufacturing Processes
Spring/Summer

535.724
Dynamics of Robots and
Spacecrafts
Spring

535.752
Advanced Flight Dynamics and
Control of Aerospace Vehicles
Fall

535.761
Hypersonic Aerothermodynamics
Spring

535.763
Aerospace Propulsion
Spring

675.722
Space Mechanical Design and
Analysis
In Development

BIOMECHANICS

10 Courses: 1 core mathematics course + 2 Group 1 courses + 3 courses from Group 1 or 2 + 4 technical electives

Group 1 Courses

535.661
Biofluid Mechanics

Fall

535.663
Biosolid Mechanics

Spring/Summer

585.631
Introduction to Biomechanics

Spring

Group 2 Courses

535.664
Fundamental Principles for Bio-
microfluidic Systems

Spring

535.667
Biomechanics of Human
Movement

Summer/Fall

535.693
Fabrication of Biomaterials,
Engineering Tissues, and Food

Fall

585.612
Biomechanical Engineering in
Sports

Fall

585.729
Cell and Tissue Engineering

Fall

585.742
Regenerative Tissue Engineering

Spring

585.743
Modeling Approaches to Cell
and Tissue Engineering

Spring

585.747
Advances in Cardiovascular
Medicine

Fall

585.762
Computational Biomechanics
and Physiological Modeling

Fall

Human Applications

535.691
Haptic Interface Design

Fall

525.786
Human Robotics Interaction

Spring

Human Robots

585.720
Orthopedic Biomechanics

Fall/Spring

535.750
Biomechanics of the cell: From
nano- and micro-mechanics to
cell organization & function

Fall

585.601
Physiology for Applied
Biomedical Engineering I

Fall/Spring

585.708
Biomaterials

Summer

585.709
Biomechanics of Cells and Stem
Cells

Fall

585.710
Biochemical Sensors

Fall/Spring

585.718
Biological Solid & Fluid
Mechanics

Summer

Other Applications

585.726
Biomimetics in Biomedical
Engineering

Summer

FLUID AND THERMAL MECHANICS

10 Courses: 1 core mathematics course + 2 Group 1 courses + 3 courses from Group 1 or 2 + 4 technical electives

Group 1 Courses

535.620
Fluid Dynamics I

Summer/Fall

535.633
Intermediate Heat Transfer

Spring

Group 2 Courses

535.625
Turbulence

Fall

535.626
Mechanics of Flight

Fall/Spring

535.670
Advanced Aerodynamics

Fall

535.761
Hypersonic Aerothermodynamics

Spring

535.763
Aerospace Propulsion

Spring

535.661
Biofluid Mechanics

Fall

535.664
Fundamental Principles for Bio-
microfluidic Systems

Spring

535.614
Fundamentals of Acoustics

Spring

535.771
Naval Architecture Design

Spring

535.773
Acoustical Oceanography

Fall

615.761
Intro to Oceanography

Fall

Aerospace Applications

Ocean Applications

535.634
Applied Heat Transfer

Summer

535.652
Thermal Systems Design and
Analysis

Fall

535.662
Energy and Environment

Spring

535.735
Computational Fluid Dynamics

Fall

Bio Applications

535.737
Multiscale Modeling and
Simulation of Mechanical
Systems

Summer

Other Applications

HYPERSONIC TECHNOLOGIES

10 Courses: 1 core mathematics course + 2 Group 1 courses + 3 courses from Group 1 or 2 + 4 technical electives

Group 1 Courses

535.608
Hypersonic Technologies and
Systems

Fall

535.734
Ultra-high Temperature Materials

Spring

535.752
Advanced Flight Dynamics and
Control of Aerospace Vehicles

Fall

535.761
Hypersonic Aerothermodynamics

Spring

Group 2 Courses

535.626
Mechanics of Flight

Fall/Spring

535.633
Intermediate Heat Transfer

Spring

535.634
Applied Heat Transfer

Summer

535.670
Advanced Aerodynamics

Fall

535.735
Computational Fluid Dynamics

Fall

Fluids/Thermal

535.671
Aerospace Materials, Structures
and Design

Fall/Spring

535.721
Advanced Composite Materials &
Manufacturing Processes

Spring

Solids and Structures

535.741
Optimal Control and
Reinforcement Learning

Spring

535.762
Guidance, Navigation and
Controls for Hypersonic Vehicles

Fall/Spring

Robotics

535.627
Computer-Aided Design

Every Term

Other
Applications

535.763
Aerospace Propulsion

Spring

Propulsion

MECHANICS OF MATERIALS AND STRUCTURES

10 Courses: 1 core mathematics course + 2 Group 1 courses + 3 courses from Group 1 or 2 + 4 technical electives

Group 1 Courses

535.632
Applied Finite Elements

Every Term

CAD/FEA

535.731
Engineering Materials: Properties
and Selections

Fall/Spring

Materials Science

535.623
Intermediate Vibrations

Fall/Spring

Dynamics/Controls

535.606
Advanced Strength of Materials

Spring

535.607
Mechanics of Solids & Structures:
Theory and Applications I

Fall

Mechanics

MECHANICS OF MATERIALS AND STRUCTURES

10 Courses: 1 core mathematics course + 2 Group 1 courses + 3 courses from Group 1 or 2 + 4 technical electives

Group 2 Courses

535.627
Computer-Aided Design

Every Term

CAD

535.663
Biosolid Mechanics

Spring/Summer

585.631
Introduction to Biomechanics

Spring

Biomechanics

535.612
Intermediate Dynamics

Fall

585.631
Introduction to Biomechanics

Spring

Dynamics

535.706
Mechanics of Solids & Structures:
Theory and Applications II

Spring

535.732
Fatigue and Fracture of Materials

Spring

535.748
Stress Waves, Impacts and
Shockwaves

Spring

535.643
Plasticity

Spring

565.604
Structural Mechanics

Fall

Mechanics

535.660
Precision Mechanical Design

Fall/Spring

565.682
Design of Ocean Structures

Summer (Odd years)

675.722
Space Mechanical Systems
Design and Analysis

In Development

Machine & Structure Design

515.602
Thermodynamics and Kinetics of
Materials

Spring

Thermodynamic

525.606
Electronic Materials

Spring

Electronic Materials

535.720
Mechanics of Composite
Materials and Structures

Fall

535.721
Advanced Composite Materials &
Manufacturing Processes

Spring/Summer

535.733
Micromechanics of
Heterogeneous Materials &
Composites

Fall

Composites

535.684
Modern Polymeric Materials

Summer/Fall

515.661
Introduction to Polymer Science

Fall

545.660
Polymer Physics

In Development

Polymer Science

535.618
Fabricatology: Advanced
Materials Processing

Summer/Fall

535.674
Smart Manufacturing and
Automation

Spring

515.658
Design for Additive
Manufacturing

Fall

Additive

535.742
Applied Machine Learning for
Mechanical Engineers

Spring/Summer

535.743
Intermediate Applied AI in
Mechanical Engineering

Fall

AI/ML

515.601
Structure and Properties of
Materials

Fall

515.617
Nanomaterials

Spring/Summer

515.622
Micro and Nano Structured
Materials & Devices

TBD

525.606
Electronic Materials

In Development

675.702
Materials for Space Systems

Every Term

Materials

535.737
Multiscale Modeling and
Simulation of Mechanical
Systems

Summer

Computation

OCEAN ENGINEERING

10 Courses: 1 core mathematics course + 2 Group 1 courses + 3 courses from Group 1 or 2 + 4 technical electives

Group 1 Courses

535.606
Advanced Strength of Materials

Spring

535.620
Fluid Dynamics I

Summer/Fall

Core ME Classes

535.771
Naval Architecture Design

Summer/Fall

565.682
Design of Ocean Structures

Summer (Odd years)

Marine Vehicles & Naval

535.773
Acoustical Oceanography

Fall

615.761
Intro to Oceanography

Fall

Ocean Sciences

Group 2 Courses

535.607
Mechanics of Solids & Structures:
Theory and Applications I

Fall

535.627
Computer-Aided Design

Every Term

535.632
Applied Finite Elements

Every Term

535.721
Advanced Composite Materials &
Manufacturing Processes

Spring/Summer

535.732
Fatigue and Fracture of Materials

Spring

535.735
Computational Fluid Dynamics

Fall

Core ME Classes

535.638
Mechanical Packaging for
Electronics Systems

Summer/Fall

535.733
Micromechanics of
Heterogeneous Materials &
Composites

Fall

535.614
Fundamentals of Acoustics

Spring

515.658
Design for Additive
Manufacturing

Fall

Marine Vehicles & Naval

615.775
Physics of Climate

Spring

Ocean Sciences

ROBOTICS, DYNAMICS, AND CONTROLS

10 Courses: 1 core mathematics course + 2 Group 1 courses + 3 courses from Group 1 or 2 + 4 technical electives

Group 1 Courses

535.642
Control Systems for Mechanical
Engineering Applications

Fall

Advanced
Control Theory

535.724
Dynamics of Robots and
Spacecraft

Spring

535.630
Kinematics & Dynamics of Robots

Fall/Spring

535.752
Advanced Flight Dynamics and
Control of Aerospace Vehicles

Fall

Advanced
Control Theory

ROBOTICS, DYNAMICS, AND CONTROLS

10 Courses: 1 core mathematics course + 2 Group 1 courses + 3 courses from Group 1 or 2 + 4 technical electives

Group 2 Courses

535.635
Introduction to Mechatronics

Spring

535.673
Mechanized Assembly: Hardware
and Algorithms

Summer

535.691
Haptic Interface Design

Fall

535.782
Haptic Applications

Spring/Summer

525.610
Microprocessors for Robotic
Systems

Spring

525.786
Human Robotics Interaction

Spring

Electronics

535.627
Computer-Aided Design

Every Term

535.612
Intermediate Dynamics

Fall

535.623
Intermediate Vibrations

Fall/Spring

535.630
Kinematics & Dynamics of Robots

Fall/Spring

535.724
Dynamics of Robots and
Spacecraft

Spring

535.628
Computer-Integrated Design &
Manufacturing

Spring

535.752
Advanced Flight Dynamics and
Control of Aerospace Vehicles

Fall

525.661
UAV Systems and Control

Fall/Spring

605.613
Introduction to Robotics

Fall/Spring

605.716
Modeling and Simulation of
Complex Systems

Spring/Summer

Dynamics

535.638
Mechanical Packaging for
Electronics Systems

Summer/Fall

535.603
Applied Optimal Control

Fall

535.622
Robot Motion Planning

Spring/Summer

535.642
Control Systems for Mechanical
Engineering Applications

Fall

535.645
Digital Control and Systems
Applications

Summer

535.659
Manufacturing Systems Analysis

Fall

535.741
Optimal Control and
Reinforcement Learning

Spring

665.681
Application of Sensing Systems

Every Term

525.609
Continuous Control Systems

Fall/Spring

525.626
Feedback Control in Biological
Signaling Pathways

Fall

535.660
Precision Mechanical Design

Fall/Spring

525.645
Modern Navigation Systems

Fall/Spring

525.777
Control Systems

Fall

Intelligence & Planning

Mechanical Design

Control Theory



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